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1 August 1980
SUPERSEDING
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19 July 1974

MILITARY SPECIFICATION

FLARING, FLANGING, AND BEADING MACHINES, TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flaring, flanging, and beading machines capable of forming single flares, double flares, and beads. The machines covered herein are also capable of forming flanges, expansions, and reductions in tubing. Type I machines are also capable of squaring and deburring tube ends prior to formation of the above tube forms.

1.2 Classification. This specification covers flaring, flanging, and beading machines of the following types, and of the sizes shown in table I. The type and size machine required for specific procurements shall be as specified (see 6.2.1).

Type I - Machines to produce single flares, flanges and beads on ferrous and non-ferrous tubing by rotary, or a combination of rotary and in-line movement, of forming tools against forming dies.

Type II- Machines to produce single and double flares, beads, reductions and expansions on ferrous and non-ferrous tubing in-line movement of forming tools against forming dies.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3441

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C113.1 - Electrical Standards for Metalworking Machine Tools (NFPA No. 79)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018).

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 4.2, 6.2.1 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition

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and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics, and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.2 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.3 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded parts and shall be removable to facilitate inspection, maintenance and repair.

All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.4 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.5 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.5 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.7 Gears. All gears shall be constructed of a material suitable for their application. Gears shall be applied in a manner to preclude either premature damage, wear or failure.

3.4 Components. All machines covered herein shall consist essentially of a base, a forming assembly, a power system, and their related control, lubrication, and electrical system.

3.4.1 Type I machines.

3.4.1.1 General principles of operation. Type I machines shall generally operate along the following principles of operation: A squared and deburred workpiece is manually inserted into a forming die until contact is made against an adjustable workpiece stop. The die is closed, securely gripping the workpiece. A rotating forming tool such as a flaring center, or beading tip, mounted onto the spindle of the machine, is manually advanced until contact is made with the workpiece and the end form completed. After completion of the forming operation, the rotating tool is returned to its starting position, the die is opened, and the workpiece removed.

3.4.1.2 Base. The machine shall have a base of adequate strength to withstand all forces applied while the machine is operating at its maximum rated capacity. The base shall mount on a workbench, base cabinet, or floor stand at a convenient operating height. When machine components are enclosed within the base, base cabinet or floor stand, removable or hinged panels shall be provided to allow access to these components for lubrication, adjustment, or maintenance.

3.4.1.3 Spindle assembly. The spindle assembly shall consist of the spindle housing, spindle sleeve, spindle bearings, spindle, spindle drive, and spindle advance mechanism. The spindle shall be mounted on ball or roller bearing contained in the spindle sleeve and the spindle sleeve in the spindle housing. The spindle shall be rotated by an electric motor through a belt and pulley drive. The spindle sleeve shall advance and retract the spindle through a gear segment and lever arrangement.

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3.4.1.3.1 Spindle. The spindle shall be sufficient in size and of the type necessary to withstand all forces encountered while operating at the machine's maximum rated capacity. The spindle end shall be constructed to mount the various beading, flanging, squaring and deburring heads and flaring centers.

3.4.1.4 Jaw assembly. The jaw assembly shall be capable of accommodating all dies necessary for single flaring and beading of tubes of the sizes specified in table I. The jaw assembly shall consist of one stationary jaw and one movable jaw with a hinging and locking mechanism. The jaws shall hold the forming dies which in turn shall firmly hold the workpiece. The stationary jaw shall be mounted to the base of the machine. The movable jaw shall hinge to the stationary jaw and shall open sufficiently to allow for insertion and removing the workpiece.

3.4.2 Type II machines. Type II machines shall be either of the single workstation type or of the multi-workstation type. On single workstation type machines, all operations shall be performed with the workpiece located at that single workstation. On multi-station machines, operations are performed by moving the workpiece from one workstation to another.

3.4.2.1 General principles of operation. Type II machines shall generally operate along the following principles: On single station, non-automatically cycled machines, a squared and deburred workpiece is manually positioned in the open die against an adjustable workpiece stop. Manual operation of the control shall close the die and grip the workpiece. Manual operation of the control to the second position shall move the workpiece stop out of position, move the forming punch into axial alignment with the tube end, and the power head shall axially force the forming punch into the tube end forming the tube end to the configuration of the punch and die. On double forming operations such as double flaring, manual operation of the control to the third position shall bring the second forming punch into axial alignment with the tube end, the power head shall axially force the second forming punch into the tube end, thereby performing the second forming operation. Moving the control to the fourth position shall move the workpiece stop back into position and shall open the die allowing the finished workpiece to be removed. On automatically sequenced machines, the squared and deburred workpiece is positioned in the open die against an adjustable workpiece stop. Actuation of the control shall initiate the automatic sequence and all operations shall be sequentially performed without further operator action. On multi-station machines, the principles of operation shall be substantially the same as with the non-automatically cycled machine except that the workpiece shall be progressively moved from one station to the next presenting the

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the workpiece to a different forming punch at each station in lieu of the forming punches moving.

3.4.2.2 Components. The machine shall consist essentially of a base, a die holder, dies, tool holder, punches, power head, a pneumatic or hydraulic system, and the necessary control system.

3.4.2.3 Base. The machine shall have a base, base cabinet, or floor stand supporting the power head at a convenient operating height. When machine components are enclosed within the base, base cabinet, or floor stand, removable or hinged panels shall be provided to allow access to these components for inspection, adjustment, and maintenance.

3.4.2.4 Die holder. The machine shall have a die holder capable of accommodating all dies necessary for single flaring, double flaring, and beading of tubes of the sizes specified in table I. The die holder shall facilitate changing of the dies and shall securely hold the die in axial alignment with the punches.

3.4.2.5 Die closing mechanism. The machine shall have a die closing mechanism that will open the die set for removal or insertion of the workpiece, and close the die set to securely grip the workpiece. The die closing mechanism shall provide adequate clamping force to prevent slippage of the workpiece during the forming operation. The mechanism shall be operated by either a hydraulic system conforming to 3.4.3 or a pneumatic system conforming to 3.4.4.

3.4.2.6 Tool holder. Single workstation machines shall have a multiposition tool holder which supports not less than two forming tools. The tool holder may be either the sliding or rotating type and shall successively align the forming tools with the forming die. In lieu of the indexable toolholder above, the machine may have multiple work stations. Multi-station machines shall have not less than three work stations, and a tool holder that holds the forming tool axial alignment with the die at each work station.

3.4.2.7 Tube stop. The machine shall have an adjustable tube stop for positioning the end of the tube in the proper position to receive the forming tool(s). The tube stop may be either mounted in the tool holder or be of the swing away type.

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3.4.2.8 Power head. The machine shall have a power head which applies axial force to the forming tools and forces the tool into the workpiece to form the end connection. The power head shall be operated by either a pneumatic system conforming to 3.4.4 or a hydraulic system conforming to 3.4.3. The power head shall exert force through a straight inline ram, an impact piston, or through a toggle arrangement, and shall be capable of applying adequate force to perform multiple forming operations on tubing of the sizes specified in table I. Means shall be provided for adjusting the amount of force exerted by the power head.

3.4.3 Hydraulic system. The hydraulic system, when furnished, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.4 Pneumatic system. The pneumatic system shall operate from an air source of 90 to 125 PSIG air pressure. The pneumatic system shall include a pressure regulator, a filtration and lubrication device, a control, and all necessary piping and hoses. The system shall be capable of withstanding an overpressure of 150 percent of operating pressure and shall be fitted with a pressure relief valve which operates at 110 percent of operating pressure.

3.4.5 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI C113.1. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI C113.1 for the disconnect fuse rating.

3.4.5.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a dripproof enclosure.

3.4.6 Tooling. Tooling shall be furnished as specified and fully described (see 6.2.1). Tooling such as punches, dies, beading tips, rollers and flaring centers shall be made to resist wear and deformation normally encountered during machine operations. The tooling shall produce the accuracies necessary

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to provide tube end connections to the dimensional accuracies specified in 3.6.1 and 3.6.2.

3.5 Machine characteristics. The machine shall meet the requirements of table I.

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TABLE I

SIZES AND CAPACITIES

CAPACITIES	TYPE I					TYPE II					
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6
Single Flaring Diameter, inches	1/8 to 2	1/8 to 3	1/2 to 4	1-1/4 to 6	1-1/4 to 8	1/8 to 3/4	1/8 to 1	1/8 to 1-1/2	1/4 to 2	1/4 to 2-1/2	1/4 to 3-1/2
Beading, Diameter, inches	1/4 to 1-1/2	1/4 to 3	5/8 to 4	1-1/4 to 6	1-1/4 to 8	1/8 to 3/4	1/8 to 1	1/4 to 1-1/2	1/4 to 1-3/4	1/4 to 2-1/8	1/4 to 3-1/4
Motor H.P. (H. P. of spindle motor or hydraulic pump motor; this entry inapplicable for Type II machines if pneumatically operated from shop supply)	1/3	3/4	2	2	2	5	7-1/2	10	10	15	15

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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3.6 Performance. Machines covered by this specification, when fitted with the proper tooling, shall be capable of meeting the requirements of 3.6.1 through 3.6.4.

3.6.1 Type I machines. Type I machines shall be capable of producing single flares, beads, and 90-degree flanges on ferrous and non-ferrous tubes of the sizes specified in table I. Type II machines shall also be capable of squaring and deburring all tubes of the above material and sizes. The accuracy of the end connections shall conform to 3.6.3. The speed of operation shall conform to 3.6.4.

3.6.2 Type II machines. Type II machines shall be capable of producing single flares, double flares, beads, and 90-degree single and double-lap flanges on ferrous and non-ferrous tube of the sizes specified in table I. The accuracy of the end connections shall conform to 3.6.3. The speed of operation shall conform to 3.6.4. Type II machines shall also be capable of forming expansions and reductions on tube ends.

3.6.3 Accuracy. The accuracy of the end fittings produced, including single flares, double flares, beads and flanges shall be square to the wall of the tube within one degree, and concentric with the outside diameter of the tube within 0.005 inch.

3.6.4 Speed of operation. When using squared and deburred tubing of Type 304 or 316 cold drawn, annealed, seamless, stainless steel tubing, the machine shall be capable of producing five samples of the same size, of any size within the range specified in table I, within the time period specified below: (The time limitations shall include workpiece handling time and actual flaring or beading time only, and shall exclude squaring and deburring time.)

Single flares - Five single flares in not more than 60 seconds

Double flares - (Type II machines only) - Five double flares in not more than 80 seconds

Beads - Five beads in not more than 80 seconds

3.7 Standard equipment. The machine shall be furnished with all standard accessory equipment such as gauges or hand tools normally furnished by the manufacturer.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

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3.9 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
U S

3.11 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination 4.4, the tests in 4.5 through 4.5.2.3, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.12.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 15 minutes, during which time the controls and all operating components shall be actuated and observed to determine that they operate properly.

4.5.2 Performance tests. Type I machines shall be subjected to the test specified in 4.5.2.1. Type II machines shall be subjected to the test specified in 4.5.2.2. Type I and type II machines shall, in addition, be subjected to the beading accuracy and speed test specified in 4.5.2.3.

4.5.2.1 Single flaring accuracy and speed (Type I machines only). Type I machines shall be tested for single flaring accuracy and speed by producing fifteen single flare samples in squared and deburred tubing meeting the requirements of AISI Type 304 or Type 316 cold drawn, annealed, seamless, stainless steel tubing. The fifteen samples shall consist of:

(a) Five samples in tubing of the smallest size specified in Table I.

(b) Five samples in tubing of the largest size specified in Table I.

(c) Five samples in tubing of a size equal to the average of the size range specified in Table I.

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During the above test, the time required to produce each 5 sample group shall be determined. All of the flares produced shall be measured for accuracy. All of the flares shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of five flares shall conform to the requirements of 3.6.4.

4.5.2.2 Double flaring accuracy and speed (Type II machines only.) Type II machines shall be tested for double flaring accuracy and speed in accordance with 4.5.2.1, except that double flares shall be produced in lieu of the single flares specified in 4.5.2.1. All of the flares shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of five flares shall conform to the requirements of 3.6.4.

4.5.2.3 Beading accuracy and speed. Type I and Type II machines shall be tested for beading accuracy and speed in accordance with 4.5.2.1, except that beads shall be produced in lieu of the single flares specified in 4.5.2.1. All of the beads shall conform to the accuracy requirements of 3.6.3. The time required to produce each group of five beads shall conform to the requirements of 3.6.4.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended for use in forming tube end connections conforming to Military Standards requirements and tolerances.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchase should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type and size machine required (see 1.2 and table I.).
c. First article approval, if required (see 3.1 and 6.3).

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(d) Specify additional safety and health requirements (see 3.2.3 and 6.4).

(e) Painting, if different (see 3.3.5).

(f) Electrical system requirements, if different (see 3.4.5).

(g) Type motor enclosure, if different (see 3.4.5.1).

(h) When tooling is required; specify type, size, and quantity and fully describe (see 3.4.6).

(i) Optional equipment, if required; fully describe (see 3.8).

(j) Lubrication plate, if required (see 3.9).

(k) Nameplate, if different (see 3.10).

(l) First article inspection, if different (see 4.2).

(m) Quality conformance inspection, if different (see 4.3).

(n) Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instruction for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instructions in all procurement instruments regarding arrangements for examinations, tests and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment: such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of individual machines in the environment) the procuring agency is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future

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operating environment. If specific point-of-operating guarding is required, the procuring agency should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(d).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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